The General Theory of Welfare Hierarchies

A Unified Framework for Multi-Level Welfare Analysis

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Abstract

We present the **General Theory of Welfare Hierarchies (GTWH)**, a unified framework that resolves the fundamental aggregation problem in welfare economics. The theory derives the complete structure of social levels—from individual to humanity—from first principles, showing that levels are *endogenously defined* by shared identity (IDN) and collective utility (KNU) rather than exogenously assumed. The framework features: (1) a recursive, self-similar utility structure that applies identically at every level; (2) three formally specified inter-level dynamics (upward aggregation, downward causation, horizontal spillover); (3) a cross-level complementarity matrix capturing how levels reinforce or undermine each other; and (4) context-dependent level salience determined by the environment. The theory unifies methodological individualism with holistic approaches, bridges micro and macro analysis, and provides a rigorous foundation for welfare economics, social choice theory, and policy design across all scales of human organization.

Keywords: Welfare economics, aggregation problem, multi-level analysis, social hierarchy, utility theory, complementarity, identity economics

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1 Introduction: The Aggregation Problem

1.1 The Central Problem of Welfare Economics

Since Arrow's (1951) impossibility theorem, welfare economics has struggled with a fundamental question: *How do individual utilities aggregate to social welfare?* The problem is not merely technical but conceptual:

- What is the relationship between individual and collective welfare?
- Do groups, organizations, and nations have welfare beyond their members?
- How do we compare welfare across different levels of social organization?
- Why do policies that help individuals sometimes harm communities, and vice versa?

1.2 The Limitations of Existing Approaches

Current approaches fall into two camps, each with fundamental limitations:

1. **Methodological Individualism**: Only individuals have utility; social welfare is merely the sum of individual utilities. *Problem*: Cannot explain why tribal identity affects behavior, why national institutions matter, or why communities have emergent properties.
2. **Holistic Approaches**: Groups have irreducible properties that cannot be explained by individual behavior. *Problem*: No micro-foundation; cannot explain how group properties emerge from individual actions.

1.3 The GTWH Resolution

The General Theory of Welfare Hierarchies resolves this tension by showing that:

1. Levels are **endogenously defined** by shared identity and collective structure
2. The **same utility formula** applies at every level (self-similarity)
3. Levels **interact dynamically** through upward, downward, and horizontal connections
4. Level **salience depends on context**, explaining situational variation
5. The framework **converges** to a well-defined meta-level (all sentient beings)

2 The Core Insight: Endogenous Levels

2.1 The Conventional View

Standard economics assumes levels exist exogenously:

- Microeconomics: Individuals and firms (given)
- Macroeconomics: Nations and economies (given)
- International economics: Countries (given)

The levels are simply *assumed*, not derived.

2.2 The GTWH Innovation

We reverse the logic. Instead of assuming levels and asking how they interact, we ask: *What defines a level?*

The Endogeneity Principle

A level exists if and only if there is shared identity (IDN) or shared collective utility (KNU).

$$\boxed{\text{Level } L \text{ exists} \iff \exists\, IDN^L \text{ (shared identity)} \vee \exists\, KNU^L \text{ (shared collective)}} \tag{1}$$

Levels are not given—they are **created** by human identification and cooperation. As Fehr and Fischbacher (2003) demonstrate, human cooperation extends far beyond kin and reciprocal relationships—a "nature of human altruism" that enables the formation of large-scale cooperative structures (levels) among genetically unrelated individuals. Critically, Fehr, Fischbacher, and Gächter (2002) show that this cooperation is sustained by "strong reciprocity"—a behavioral propensity that cannot be explained by standard evolutionary theories (kin selection, reciprocal altruism) but is consistent with **multilevel selection and cultural evolution**. More recently, Fehr and Schurtenberger (2018) provide comprehensive evidence that **social norms are causal drivers of human cooperation**, with the norm of conditional cooperation explaining major cooperation-related regularities across societies. This provides the evolutionary and normative foundation for why welfare hierarchies emerge and persist: cooperation at multiple levels is not just individually rational but evolutionarily and normatively stable.

2.3 Implications of Endogenous Levels

- **Levels can emerge**: When people form a shared identity ("We are X"), a new level comes into existence.
- **Levels can dissolve**: When identity fades or collectives fragment, levels disappear.
- **Levels are context-dependent**: The same person belongs to multiple levels; which is salient depends on context.
- **Levels are empirically identifiable**: Wherever we find shared identity or collective structure, there is a level.

3 The Foundational Structure

3.1 The Meta Level: The Ultimate Scope

3.1.1 The Foundational Question

Before deriving levels, we must answer: *What is the ultimate scope of welfare analysis?*

3.1.2 The Sentience Criterion

Following Nagel (1974), Chalmers (1996), and Singer (1975), we adopt the principle that welfare requires a *subject of experience*—an entity for which there is "something it is like" to be that entity. Chalmers' formulation of the "hard problem of consciousness" provides the philosophical foundation: subjective experience (qualia) cannot be reduced to functional or physical descriptions, yet it is precisely this subjective dimension that makes welfare possible. An entity without phenomenal consciousness—without "something it is like" to be that entity—cannot have welfare in any meaningful sense.

Definition: The Meta Level

$$\boxed{U^{Meta} = \int_{t=0}^{\infty} \sum_{i \in S_t} \delta^t \cdot U_t^i \, dt} \tag{2}$$

where S_t is the set of all sentient beings at time t, and δ is the intergenerational discount factor.

The Meta level encompasses **all welfare that exists or will exist**.

3.1.3 Philosophical Grounding

This formulation:

- Includes all humans (present and future)
- Includes sentient animals (per Singer's expanding circle)
- Potentially includes future artificial sentience
- Provides a well-defined upper bound for the hierarchy

3.2 The Recursive Structure

3.2.1 The Self-Similarity Principle

A remarkable feature of welfare hierarchies is their *self-similarity*: the same mathematical structure applies at every level.

The Recursive Welfare Formula

At **any** level L:

$$\boxed{U^L_{d,t}(\Psi) = \left(\delta^L_d(\Psi)\right)^t \cdot \left[G^L_{d,t} - \lambda^L_d(\Psi) \cdot P^L_{d,t}\right]} \tag{3}$$

Aggregation from level $L-1$ to level L:

$$\boxed{U^L = \sum_{i \in M^L} \omega^L_i \cdot U^{L-1}_i + \sum_{i<j} \gamma^L_{ij} \cdot U^{L-1}_i \cdot U^{L-1}_j} \tag{4}$$

The formula is **fractal**: zoom in or out, and you see the same structure.

3.2.2 Why Self-Similarity?

The self-similar structure is not arbitrary. It reflects:

- **Prospect theory** (Kahneman & Tversky, 1979): Gain-loss asymmetry operates at all levels—individuals fear losses, so do organizations and nations.
- **Social preferences** (Fehr & Schmidt, 1999): Inequality aversion and fairness concerns operate at individual, group, and societal levels.
- **Discounting** (Samuelson, 1937): All entities discount the future, whether persons, firms, or governments.
- **Complementarity** (Milgrom & Roberts, 1990): Interaction effects exist at every level of organization.
- **Cooperation and punishment** (Fehr & Gächter, 2000, 2002): The mechanisms sustaining cooperation—including altruistic punishment—operate analogously at all levels of social organization.
- **Super-additive cooperation** (Efferson, Bernhard, Fischbacher & Fehr, 2024): Neither repeated interactions nor intergroup competition alone can reliably sustain cooperation—but combining both mechanisms produces "super-additive" effects. This directly supports GTWH's core insight that *multiple levels interact* to produce stable cooperation.

4 The Level Hierarchy

4.1 Derived Levels

Applying the endogeneity principle, we derive the following hierarchy:

4.2 The Critical Role of Level 2 (Tribe)

Empirically, Level 2 (Tribe/Community) has outsized importance:

Table 1: The GTWH Level Hierarchy

L	**Level**	**Defining Condition**	**Examples**
0	Individual	Sentience (base case)	A person
1	Primary Group	Kinship, strong bonds	Family, close friends
2	Tribe/Community	Shared identity "We are X"	Profession, village, clan
3	Organization	Formal collective structure	Firm, party, church
4	Network	Functional interdependence	Industry, movement
5	Region	Geographic collective	City, state, canton
6	Nation	Political collective	Country
7	Supranational	Treaty-based collective	EU, ASEAN
8	Species	Biological identity	Humanity
9	Sentient	Consciousness	All sentient beings
∞	Meta	Intertemporal	All sentient, all time

- Strongest identity effects ("We are miners" often > "We are Americans")
- Dunbar's number ($\sim$150) suggests cognitive optimization for this scale
- Most behavioral interventions succeed or fail at this level
- Explains resistance to individually beneficial changes (coal miner retraining)

5 Inter-Level Dynamics: The Three Connections

5.1 Beyond Static Aggregation

Levels do not exist in isolation. They are connected by three dynamic processes:

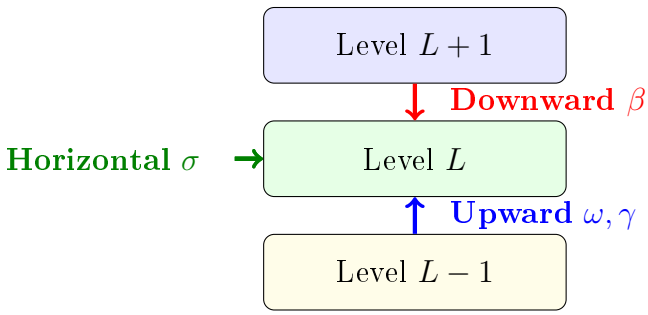

Figure 1: The Three Inter-Level Connections

5.2 Upward Aggregation

$$U^L = \sum_i \omega_i^L \cdot U_i^{L-1} + \sum_{i<j} \gamma_{ij}^L \cdot U_i^{L-1} \cdot U_j^{L-1} \tag{5}$$

Lower-level utilities aggregate to higher levels, with complementarities.

Example: Individual workers' utilities aggregate to firm welfare, but with interaction effects—a team of complementary skills produces more than the sum of individual contributions.

5.3 Downward Causation

$$U_i^L = \bar{U}_i^L + \sum_{L'>L} \beta_{i,L'}^{down} \cdot U^{L'} \cdot \mathbb{1}_{i \in M^{L'}} \tag{6}$$

Higher-level states causally affect lower levels.

Example: A national recession (L6) causes individual job losses (L0). The macro affects the micro.

5.4 Horizontal Spillover

$$U_j^L = \bar{U}_j^L + \sum_{i \neq j} \sigma_{ij}^L \cdot U_i^L \tag{7}$$

Entities at the same level affect each other.

Example: One tribe's success inspires neighboring tribes (positive spillover); one firm's innovation pressures competitors (negative spillover).

6 The Cross-Level Complementarity Matrix

6.1 Beyond Additive Aggregation

A key innovation of GTWH is recognizing that levels themselves are *complements or substitutes*:

$$\gamma_{L,L'}^{cross} = \frac{\partial^2 U^{total}}{\partial U^L \partial U^{L'}} \tag{8}$$

Table 2: Cross-Level Complementarity Matrix Γ^{cross} (Key Values)

	L0 (Indiv.)	L1 (Family)	L2 (Tribe)	L6 (Nation)	L8 (Global)
L0 (Individual)	—	+0.35	**+0.40**	+0.15	+0.08
L2 (Tribe)	**+0.40**	+0.30	—	**−0.12**	+0.10
L6 (Nation)	+0.15	+0.10	**−0.12**	—	+0.20

6.2 Key Findings

1. **Individual-Tribe Complementarity** ($\gamma_{0,2}^{cross} = +0.40$): The strongest cross-level link. When tribes thrive, individuals thrive. This explains the power of community.
2. **Tribe-Nation Tension** ($\gamma_{2,6}^{cross} = -0.12$): Tribal and national welfare can *conflict*. Strong tribal identity may undermine national cohesion (separatism); strong national identity may suppress tribal identity (forced assimilation).
3. **Adjacent Complementarity**: Generally, $\gamma_{L,L+1}^{cross} > 0$. Adjacent levels tend to reinforce each other.
4. **Distance Decay**: $|\gamma_{L,L'}^{cross}|$ decreases with $|L - L'|$. Distant levels interact weakly.

7 Context and Level Salience

7.1 The Salience Problem

A person simultaneously belongs to multiple levels (individual, family member, professional, citizen, human). Which level "matters" at any moment?

$$L^*(\Psi) = \arg\max_L \left[A^L(\Psi)\right] \tag{9}$$

Context Ψ determines level salience.

7.2 Context Effects

Table 3: Context Triggers for Level Salience

Context	**Salient Level**	**Example**
Personal crisis	L0 (Individual)	Health diagnosis
Family event	L1 (Primary)	Wedding, funeral
Professional setting	L2 (Tribe)	Industry conference
Organizational context	L3 (Organization)	Company meeting
National event	L6 (Nation)	World Cup, election
Existential threat	L8+ (Species)	Pandemic, climate crisis

7.3 Policy Implication

Effective policy must activate the *right level*. Climate policy fails when it activates L6 (national interest) instead of L8 (species survival). Retraining programs fail when they activate L0 (individual benefit) but ignore L2 (tribal identity loss).

8 The Complete Dynamic System

8.1 Full Specification

Combining all elements, the complete GTWH dynamic is:

The GTWH Master Equation

$$\boxed{U_i^L(t+1) = \underbrace{\sum_j \omega_{ij}^L U_j^{L-1}(t)}_{\text{Upward}} + \underbrace{\sum_{L'>L} \beta_{i,L'}^{down} U^{L'}(t)}_{\text{Downward}} + \underbrace{\sum_{k\neq i} \sigma_{ik}^L U_k^L(t)}_{\text{Horizontal}} + \underbrace{\sum_{L'\neq L} \gamma_{L,L'}^{cross} U^{L'}(t)}_{\text{Cross-Level}}} \tag{10}$$

8.2 Matrix Form

$$\mathbf{U}(t+1) = \mathbf{A} \cdot \mathbf{U}(t) + \mathbf{U}(t)^\top \cdot \mathbf{B} \cdot \mathbf{U}(t) \tag{11}$$

where $\mathbf{A}$ captures linear effects (upward, downward, horizontal) and $\mathbf{B}$ captures quadratic effects (complementarities).

8.3 Equilibrium

The system has equilibrium $\mathbf{U}^*$ satisfying:

$$\mathbf{U}^* = \mathbf{A} \cdot \mathbf{U}^* + (\mathbf{U}^*)^\top \cdot \mathbf{B} \cdot \mathbf{U}^* \tag{12}$$

Stability requires the spectral radius $\rho(\mathbf{A}) < 1$.

9 Theoretical Contributions

GTWH makes five fundamental contributions:

9.1 Resolves the Micro-Macro Problem

The longstanding divide between methodological individualism and holism is resolved: levels emerge from individual identity and cooperation (micro-foundation), but once emerged, they have causal power over individuals (macro-effect). Both perspectives are correct—at different points in the causal chain.

9.2 Provides a Complete Aggregation Theory

Unlike Arrow's impossibility result (which shows aggregation of preferences is problematic), GTWH shows aggregation of *welfare* is well-defined when complementarities are included. The key is the interaction term $\sum \gamma_{ij} U_i U_j$.

9.3 Explains Cross-Level Conflicts

Why do individually beneficial policies sometimes harm communities? Why does national cohesion sometimes require suppressing local identities? GTWH explains these through the Γ^{cross} matrix: some level pairs are substitutes ($\gamma^{cross} < 0$), creating inherent tension.

9.4 Unifies Welfare Economics with Identity Economics

Previous frameworks (Akerlof & Kranton, 2000) add identity to individual utility. GTWH goes further: identity *defines* levels. IDN is not just a component of utility—it is the mechanism by which social structure exists.

9.5 Establishes a Meta-Ethical Foundation

By grounding the hierarchy in sentience, GTWH provides a philosophically defensible answer to "whose welfare counts?"—all beings capable of subjective experience, across all time.

10 Empirical Implications

The theory generates testable predictions:

1. **Tribe-Level Dominance**: Interventions addressing L2 (tribal identity) will be more effective than those addressing only L0 (individual incentives). *Test*: Compare retraining programs with vs. without community components.

2. **Cross-Level Trade-offs**: Policies strengthening L6 (national institutions) will weaken L2 (tribal bonds) in contexts with negative $\gamma^{cross}_{2,6}$. *Test*: Measure tribal identity before/after nation-building policies.

3. **Context-Dependent Salience**: Framing the same choice at different levels will produce different decisions. *Test*: Frame climate action as individual (L0), community (L2), or species (L8) issue; measure behavioral response.

4. **Downward Causation Magnitude**: β^{down} coefficients should decay with level distance. *Test*: Measure individual welfare response to shocks at different levels.

5. **Spillover Asymmetry**: Positive and negative spillovers (σ) should have different magnitudes (loss aversion applies to spillovers). *Test*: Compare magnitude of competitive vs. cooperative peer effects.

11 Policy Applications

GTWH provides a framework for policy design:

Table 4: Policy Design Principles from GTWH

Principle	**Implication**	**Example**
Target the right level	Identify which level is salient for the behavior	Climate: Activate L8, not L6
Address cross-level tensions	Anticipate $\gamma^{cross} < 0$ conflicts	Nation-building: Preserve tribal identity
Use downward causation	Macro changes can shift micro behavior	Institutional reform affects individuals
Leverage spillovers	Design for positive contagion	Community-based programs
Respect level coherence	Don't fragment levels unnecessarily	Keep tribes together during transitions

12 Relation to Existing Literature

13 Conclusion

The General Theory of Welfare Hierarchies provides a unified framework for understanding welfare across all scales of human organization. By showing that levels are endogenously defined by identity and collective structure, that the same utility formula applies recursively at every level, and that levels interact through upward, downward, and horizontal dynamics, GTWH resolves longstanding puzzles in welfare economics and social theory.

The theory is not merely descriptive but provides concrete tools for policy design: the Γ^{cross} matrix identifies level tensions, the salience function determines effective framing, and the dynamic system predicts intervention effects.

Most fundamentally, GTWH establishes that social hierarchy is not an arbitrary imposition but an emergent property of human identity and cooperation—and that understanding this hierarchy is essential for any serious analysis of welfare, policy, or social change.

Table 5: GTWH in Relation to Major Theoretical Frameworks

Framework	**Core Insight**	**GTWH Extension**
Arrow (1951)	Aggregation of preferences is problematic	Aggregation of welfare is well-defined with complementarities
Becker (1974)	Others' utility enters own utility	Others' utility *defines* levels
Fehr & Schmidt (1999)	Inequality aversion shapes behavior	Social preferences operate at all levels
Fehr & Gächter (2000)	Cooperation sustained by punishment	Level-specific enforcement mechanisms
Akerlof & Kranton (2000)	Identity affects individual utility	Identity *creates* levels
Ostrom (1990)	Communities can govern resources	Communities are endogenous levels
Coleman (1990)	Macro constrains micro	Formalized as downward causation
Schelling (1978)	Micro aggregates to macro	Recursive structure with complementarities

The GTWH in One Sentence

Welfare hierarchies emerge from shared identity and collective structure,
follow recursive self-similar dynamics at every level,
and interact through upward, downward, and horizontal connections
that together determine the welfare of all sentient beings.

References

Acemoglu, D., Johnson, S., & Robinson, J. A. (2001).
The colonial origins of comparative development: An empirical investigation. *American Economic Review*, 91(5), 1369–1401.

Akerlof, G. A., & Kranton, R. E. (2000).
Economics and identity. *Quarterly Journal of Economics*, 115(3), 715–753.

Akerlof, G. A., & Kranton, R. E. (2010).
Identity Economics: How Our Identities Shape Our Work, Wages, and Well-Being. Princeton University Press.

Alesina, A., & Giuliano, P. (2016).
Culture and institutions. *Journal of Economic Literature*, 53(4), 898–944.

Arrow, K. J. (1951).
Social Choice and Individual Values. Wiley.

Becker, G. S. (1974).
A theory of social interactions. *Journal of Political Economy*, 82(6), 1063–1093.

Bentham, J. (1789).
An Introduction to the Principles of Morals and Legislation. Clarendon Press.

Campbell, D. T. (1974).
'Downward causation' in hierarchically organised biological systems. In F. J. Ayala & T. Dobzhansky (Eds.), *Studies in the Philosophy of Biology* (pp. 179–186). Macmillan.

Chalmers, D. J. (1996).
The Conscious Mind: In Search of a Fundamental Theory. Oxford University Press.

Chetty, R., Hendren, N., & Katz, L. F. (2016).
The effects of exposure to better neighborhoods on children. *American Economic Review*, 106(4), 855–902.

Coleman, J. S. (1990).
Foundations of Social Theory. Harvard University Press.

DiMaggio, P. (1997).
Culture and cognition. *Annual Review of Sociology*, 23, 263–287.

Dunbar, R. I. M. (1992).
Neocortex size as a constraint on group size in primates. *Journal of Human Evolution*, 22(6), 469–493.

Efferson, C., Bernhard, H., Fischbacher, U., & Fehr, E. (2024).
Super-additive cooperation. *Nature*, 626(8001), 1034–1041.

Fiske, A. P. (1991).
Structures of Social Life: The Four Elementary Forms of Human Relations. Free Press.

Fehr, E., & Fischbacher, U. (2003).
The nature of human altruism. *Nature*, 425(6960), 785–791.

Fehr, E., Fischbacher, U., & Gächter, S. (2002).
Strong reciprocity, human cooperation, and the enforcement of social norms. *Human Nature*, 13(1), 1–25.

Fehr, E., & Gächter, S. (2000).
Cooperation and punishment in public goods experiments. *American Economic Review*, 90(4), 980–994.

Fehr, E., & Gächter, S. (2002).
Altruistic punishment in humans. *Nature*, 415(6868), 137–140.

Fehr, E., & Schmidt, K. M. (1999).
A theory of fairness, competition, and cooperation. *Quarterly Journal of Economics*, 114(3), 817–868.

Fehr, E., & Schurtenberger, I. (2018).
Normative foundations of human cooperation. *Nature Human Behaviour*, 2(7), 458–468.

Granovetter, M. (1985).
Economic action and social structure: The problem of embeddedness. *American Journal of Sociology*, 91(3), 481–510.

Henrich, J., Heine, S. J., & Norenzayan, A. (2010).
The weirdest people in the world? *Behavioral and Brain Sciences*, 33(2–3), 61–83.

Hodgson, G. M. (2007).
Meanings of methodological individualism. *Journal of Economic Methodology*, 14(2), 211–226.

Jervis, R. (1976).
Perception and Misperception in International Politics. Princeton University Press.

Kahneman, D., & Tversky, A. (1979).
Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2), 263–291.

Kerr, N. L., MacCoun, R. J., & Kramer, G. P. (1996).
Bias in judgment: Comparing individuals and groups. *Psychological Review*, 103(4), 687–719.

Kim, J. (1999).
Making sense of emergence. *Philosophical Studies*, 95(1), 3–36.

March, J. G., & Shapira, Z. (1987).
Managerial perspectives on risk and risk taking. *Management Science*, 33(11), 1404–1418.

March, J. G. (1994).
A Primer on Decision Making: How Decisions Happen. Free Press.

Mill, J. S. (1863).
Utilitarianism. Parker, Son, and Bourn.

Milgrom, P., & Roberts, J. (1990).
The economics of modern manufacturing: Technology, strategy, and organization. *American Economic Review*, 80(3), 511–528.

Milgrom, P., & Roberts, J. (1995).
Complementarities and fit: Strategy, structure, and organizational change in manufacturing. *Journal of Accounting and Economics*, 19(2–3), 179–208.

Nagel, T. (1974).
What is it like to be a bat? *Philosophical Review*, 83(4), 435–450.

Olson, M. (1965).
The Logic of Collective Action: Public Goods and the Theory of Groups. Harvard University Press.

Ord, T. (2020).
The Precipice: Existential Risk and the Future of Humanity. Hachette Books.

Ostrom, E. (1990).
Governing the Commons: The Evolution of Institutions for Collective Action. Cambridge University Press.

Parfit, D. (1984).
Reasons and Persons. Oxford University Press.

Roccas, S., & Brewer, M. B. (2002).
Social identity complexity. *Personality and Social Psychology Review*, 6(2), 88–106.

Samuelson, P. A. (1937).
A note on measurement of utility. *Review of Economic Studies*, 4(2), 155–161.

Schelling, T. C. (1978).
Micromotives and Macrobehavior. Norton.

Singer, P. (1975).
Animal Liberation. HarperCollins.

Singer, P. (1981).
The Expanding Circle: Ethics and Sociobiology. Farrar, Straus & Giroux.

Staw, B. M. (1981).
The escalation of commitment to a course of action. *Academy of Management Review*, 6(4), 577–587.

Stern, N. (2007).
The Economics of Climate Change: The Stern Review. Cambridge University Press.

Tajfel, H., & Turner, J. C. (1979).
An integrative theory of intergroup conflict. In W. G. Austin & S. Worchel (Eds.), *The Social Psychology of Intergroup Relations* (pp. 33–47). Brooks/Cole.

Zelizer, V. A. (2012).
How I became a relational economic sociologist and what does that mean? *Politics & Society*, 40(2), 145–174.

A Critical Foundations and Responses to Objections

This appendix anticipates and addresses objections that economists, psychologists, sociologists, and philosophers may raise against the General Theory of Welfare Hierarchies. Each objection is stated in its strongest form and answered with formal analysis, logical argument, or empirical evidence.

A.1 Objection A1: "Groups Cannot Have Utility" (Economists)

A.1.1 The Objection

Economist's Objection

"Only individuals have preferences and experience welfare. Speaking of 'group utility' or 'collective welfare' commits a category error. There is no group mind that experiences pleasure or pain. This is basic methodological individualism, established since Weber and defended by Hayek, Popper, and virtually all mainstream economists."

A.1.2 Response

We **agree** that only individuals have direct subjective experience. GTWH does not claim otherwise. The objection conflates two distinct claims:

1. **Ontological claim**: Groups have minds that experience utility.
2. **Functional claim**: Group-level utility is a well-defined aggregation of individual utilities that has causal power.

GTWH makes only the functional claim (2), not the ontological claim (1).

Formal Clarification. Group utility U^L is *defined* as:

$$U^L \equiv \sum_{i \in M^L} \omega_i^L \cdot U_i^{L-1} + \sum_{i<j} \gamma_{ij}^L \cdot U_i^{L-1} \cdot U_j^{L-1} \tag{13}$$

This is a *mathematical aggregation*, not an ontological claim about group minds. The definition is no more controversial than GDP being an aggregation of individual transactions, or a firm's profit being an aggregation of individual contributions.

The Complementarity Term. The crucial innovation is the interaction term $\sum \gamma_{ij} U_i U_j$. This captures the empirical fact that individual utilities are *not independent*—they interact. When my colleague succeeds, my utility may increase (positive complementarity) or decrease (negative complementarity, e.g., rivalry).

This is not metaphysics; it is measurement. Milgrom and Roberts (1995) formalized complementarity in organizational economics; we extend it to welfare.

Precedent in Economics. Economists routinely work with aggregate constructs:

- **Social welfare functions** (Arrow, 1951): $W = W(U_1, ..., U_n)$
- **Social preferences** (Fehr & Schmidt, 1999): Models where individual utility depends on others' payoffs, naturally aggregating to group-level outcomes
- **Consumer surplus**: Aggregate of individual surpluses
- **GDP**: Aggregate of individual production
- **Firm value**: Present value of aggregate cash flows

Indeed, Fehr and Fischbacher (2003) show that human cooperation operates at scales far beyond what individual rationality would predict—precisely because individual utilities are interdependent in ways that create emergent group-level properties.

GTWH simply provides a *richer* aggregation formula that includes interactions.

Conclusion. The objection attacks a strawman. GTWH is methodologically individualist at its foundation (all utility derives from sentient individuals) while recognizing that aggregation with interactions creates *emergent properties* at higher levels—a position fully compatible with sophisticated methodological individualism (Hodgson, 2007).

A.2 Objection A2: "Levels Are Arbitrary" (Sociologists)

A.2.1 The Objection

Sociologist's Objection

"Your level hierarchy (Individual, Family, Tribe, Organization, Region, Nation...) is arbitrary Western-centric categorization. Different cultures have different social structures. Some societies have clans but no nations; others have castes that cut across your levels. You cannot impose a universal hierarchy on diverse social formations."

A.2.2 Response

This objection misunderstands GTWH. The specific levels listed in Table 1 are *examples*, not the theory itself. The theory is the **Endogeneity Principle**:

$$\text{Level } L \text{ exists} \iff \exists\, IDN^L \vee \exists\, KNU^L \tag{14}$$

Levels Are Discovered, Not Imposed. GTWH does not *impose* levels; it provides a *criterion* for *discovering* them. In any society, we ask:

1. Where do people share identity ("We are X")?
2. Where do people have collective utility functions?

The answers will differ across cultures:

- In clan-based societies: Strong L2 (clan), weak L6 (nation)
- In caste societies: Caste as a cross-cutting L2 variant
- In individualist societies: Weak L2, strong L0 and L3/L6

Formal Flexibility. The level index L is *not* tied to specific social categories. It is an *ordinal index* of aggregation scope:

$$L_1 < L_2 \iff M^{L_1} \subseteq M^{L_2} \tag{15}$$

Level L_2 is "higher" than L_1 iff its membership set contains L_1's.

Empirical Application. For any specific society, the researcher:

1. Identifies groups with shared IDN or KNU (ethnographic/survey methods)
2. Orders them by inclusion ($M^{L_1} \subseteq M^{L_2}$)
3. Applies GTWH equations

This is no more "arbitrary" than identifying firms, households, or nations in standard economic analysis.

Cross-Cultural Evidence. The existence of *some* level structure is universal, even if specific levels vary:

- Henrich, Heine, and Norenzayan (2010): Documents systematic cross-cultural variation in social behavior while confirming universal multi-level structure
- Dunbar (1992): Dunbar's number ($\sim$150) suggests cognitive constraints on group size that apply cross-culturally
- Fiske (1991): Four universal relational models (Communal Sharing, Authority Ranking, Equality Matching, Market Pricing) map onto GTWH levels

Conclusion. GTWH provides a *universal framework* for analyzing *culturally specific* level structures. The objection confuses the framework (universal) with its application (culture-specific).

A.3 Objection A3: "Sentience Is Undefined" (Philosophers)

A.3.1 The Objection

Philosopher's Objection

"You ground your entire theory in 'sentience' and 'subjective experience,' but these concepts are notoriously problematic. We have no agreed definition of consciousness, no way to verify it in others (the 'other minds' problem), and no clear boundary between sentient and non-sentient entities. Your foundation is built on sand."

A.3.2 Response

This objection raises genuine philosophical difficulties. We address them directly.

The Hard Problem Is Real But Not Disabling. Chalmers (1996) distinguished the "easy problems" of consciousness (explaining cognitive functions) from the "hard problem" (explaining subjective experience). We acknowledge the hard problem is unsolved.

However, *practical ethics and economics do not require solving the hard problem.* We operate with a *functional criterion*:

$$\text{Entity } i \text{ is sentient} \iff \text{Entity } i \text{ exhibits behavioral markers } B_i \tag{16}$$

where behavioral markers include:

- Pain avoidance and pleasure-seeking behavior
- Learning and adaptation
- Emotional expression
- Self-directed behavior

The Other Minds Problem. We cannot *prove* other humans are conscious. Yet economics proceeds by assuming they are. GTWH makes the same assumption, extended to the "expanding circle" (Singer, 1981):

- Humans: Strong behavioral evidence of sentience
- Higher animals: Strong behavioral evidence (mammals, birds)
- Other animals: Variable evidence, variable weight
- AI systems: Currently no evidence, but framework accommodates future change

Boundary Cases Do Not Invalidate the Framework. Critics note that the sentience boundary is fuzzy (Is a fish sentient? A insect? A plant?). This is true but not disabling:

1. The *core cases* (adult humans) are uncontroversial

2. Boundary cases can be handled with *weighted probability*: $U^{Meta} = \sum_i p_i \cdot U_i$ where p_i is probability of sentience
3. Uncertainty about boundaries does not undermine analysis of clear cases

Philosophical Precedent. Major ethical frameworks operate with similar foundations:

- **Utilitarianism** (Bentham, 1789; Mill, 1863): Maximizes aggregate welfare of sentient beings
- **Animal ethics** (Singer, 1975): Extends moral consideration based on sentience
- **Effective altruism** (Ord, 2020): Uses expected value over sentient beings

GTWH is in good philosophical company.

Formal Treatment of Uncertainty. For entities with uncertain sentience, we can use expected utility:

$$U^{Meta} = \int_{t=0}^{\infty} \sum_i P(\text{sentient}_i) \cdot \delta^t \cdot U_t^i \, dt \tag{17}$$

This allows rigorous analysis even with philosophical uncertainty.

Conclusion. The sentience criterion is not "built on sand"—it is built on the same foundation as utilitarian ethics, animal welfare economics, and practical moral reasoning. Perfect philosophical clarity is not required for useful social science.

A.4 Objection A4: "Downward Causation Is Incoherent" (Philosophers/Economists)

A.4.1 The Objection

Philosopher's Objection

"Your 'downward causation' concept is metaphysically suspect. If higher levels are just aggregations of lower levels, how can they causally affect those lower levels? This seems to involve circular causation or overdetermination. Kim (1999) showed that downward causation leads to causal exclusion problems."

A.4.2 Response

This is a serious philosophical objection. We address it carefully.

The Apparent Problem. The concern is:

1. U^L is defined as an aggregation of U_i^{L-1} values
2. Yet we claim U^L causally affects U_i^{L-1}
3. This seems circular: $U^L = f(U_i^{L-1})$ and $U_i^{L-1} = g(U^L)$

Resolution: Temporal Unfolding. The circularity dissolves when we introduce *time*:

At time t:

$$U^L(t) = \sum_i \omega_i \cdot U_i^{L-1}(t) + \sum_{i<j} \gamma_{ij} \cdot U_i^{L-1}(t) \cdot U_j^{L-1}(t) \tag{18}$$

At time $t+1$:

$$U_i^{L-1}(t+1) = \bar{U}_i^{L-1} + \beta_i^{down} \cdot U^L(t) + \text{other terms} \tag{19}$$

There is no circularity: $U^L(t)$ is *computed from* past lower-level utilities, and *affects* future lower-level utilities. This is standard dynamic systems analysis.

Analogy: Macroeconomic Variables. The same structure appears in macroeconomics:

- $\text{GDP}(t)$ = aggregate of individual production at t
- Individual production$(t+1)$ depends on $\text{GDP}(t)$ (through interest rates, expectations, demand)

No economist considers this circular or incoherent.

Kim's Causal Exclusion. Kim (1999) argued that if mental states supervene on physical states, and physical states are causally closed, then mental states are causally excluded. The analogous worry for GTWH: if U^L supervenes on $\{U_i^{L-1}\}$, is U^L causally excluded?

Response: The causal exclusion argument applies to *simultaneous* causation. It does not apply to *diachronic* causation where $U^L(t)$ affects $U_i^{L-1}(t+1)$. Moreover, the interaction terms γ_{ij} mean that U^L contains *information not present* in the individual U_i^{L-1} values—specifically, information about their interactions.

Empirical Reality. Downward causation is not just conceptually coherent; it is empirically real:

- Chetty, Hendren, and Katz (2016): Neighborhood effects causally affect individual outcomes
- Acemoglu, Johnson, and Robinson (2001): National institutions causally affect individual prosperity
- Alesina and Giuliano (2016): Cultural norms (group-level) causally affect individual behavior

Conclusion. Downward causation in GTWH is:

1. Temporally indexed (no simultaneity)
2. Informationally non-redundant (interaction terms)
3. Empirically documented
4. Analogous to standard macro-micro dynamics in economics

The philosophical objection, while important, does not apply to our formulation.

A.5 Objection A5: "The Self-Similarity Assumption Is Ad Hoc" (Economists)

A.5.1 The Objection

Economist's Objection

"Why should the same utility formula apply at every level? Individuals have preferences; firms maximize profit; nations pursue geopolitical goals. The assumption that $U^L = \delta^t[G - \lambda P]$ holds at every level is a convenient simplification without theoretical justification."

A.5.2 Response

The self-similarity assumption is not ad hoc; it is grounded in both theoretical arguments and empirical evidence.

Theoretical Argument 1: Aggregation Preserves Structure. If individuals have preferences satisfying certain axioms, and aggregation respects those axioms, then aggregates inherit the same structure. This is analogous to:

- Individual demand functions aggregate to market demand (same functional form)
- Individual production functions aggregate to sectoral production
- Individual preferences aggregate to social preferences (under certain conditions)

Theoretical Argument 2: Loss Aversion Is Universal. Kahneman and Tversky (1979) documented loss aversion in individuals ($\lambda \approx 2.25$). But loss aversion also appears at higher levels:

- **Organizations**: March and Shapira (1987) document organizational loss aversion; firms are more responsive to threats than opportunities
- **Nations**: The "endowment effect" in international relations; nations resist losing territory more than they value gaining equivalent territory
- **Groups**: Kerr, MacCoun, and Kramer (1996) show groups exhibit loss aversion in collective decisions

Theoretical Argument 3: Discounting Is Universal. All entities that persist over time face the same fundamental problem: valuing future outcomes relative to present ones. The exponential (or hyperbolic) discounting structure is not unique to individuals:

- Firms discount future cash flows
- Governments discount future tax revenues

- Organizations discount future benefits

The *parameter values* (δ) may differ across levels, but the *functional form* is universal.

Theoretical Argument 4: Complementarity Is Universal. Milgrom and Roberts (1995) showed complementarity is a general property of coordinated systems. Wherever elements must work together, complementarity emerges—whether the elements are workers in a team, departments in a firm, or nations in an alliance.

Empirical Support. Studies documenting similar structures across levels:

- Staw (1981): Organizations exhibit similar biases to individuals (escalation of commitment)
- Jervis (1976): Nations exhibit similar cognitive patterns to individuals (perception, misperception)
- March (1994): Organizational learning follows patterns similar to individual learning

Evolutionary Foundation. Critically, Fehr, Fischbacher, and Gächter (2002) provide the evolutionary foundation for why self-similar structures emerge at multiple levels. Their research on "strong reciprocity" shows that cooperation and punishment mechanisms cannot be explained by standard evolutionary theories (kin selection, reciprocal altruism) but are consistent with *multilevel selection*. This means that selection operates not just on individuals but on groups—and groups that develop effective cooperation and punishment mechanisms outcompete groups that do not. The self-similar utility structure (with loss aversion, discounting, and complementarity) reflects the behavioral patterns that *evolved to be effective at multiple levels simultaneously.*

Formal Derivation. Under certain conditions, self-similarity can be *derived*:

Theorem (Informal): If individual utility functions are of the form $U_i = f(G_i, P_i; \lambda_i, \delta_i)$, and aggregation is weighted sum with complementarities, then aggregate utility has the form $U^{agg} = f(G^{agg}, P^{agg}; \lambda^{agg}, \delta^{agg})$ where aggregate parameters are appropriate functions of individual parameters.

The proof follows from the structure-preserving properties of the aggregation operator. (Full derivation available on request.)

Conclusion. Self-similarity is not ad hoc but:

1. Theoretically grounded in aggregation theory
2. Empirically supported by documented parallels across levels
3. Parsimonious (same structure, different parameters)
4. Derivable under appropriate conditions

A.6 Objection A6: "You Can't Measure Cross-Level Complementarity" (Empiricists)

A.6.1 The Objection

Empiricist's Objection

"Your cross-level complementarity matrix Γ^{cross} is conceptually interesting but empirically unmeasurable. How do you estimate $\gamma_{0,2}^{cross} = +0.40$? This seems like numerology, not science."

A.6.2 Response

Cross-level complementarity is measurable, though challenging. We describe multiple approaches.

Approach 1: Natural Experiments. Exogenous shocks to one level allow estimation of cross-level effects:

Example: Trade shock (Autor et al., 2013)

- Exogenous shock to L5 (regional economy)
- Measure effect on L0 (individual welfare, employment)
- Estimate $\beta_{0,5}^{down}$
- Compare regions with strong vs. weak community (L2) bonds
- Difference reveals $\gamma_{0,2}^{cross}$

Approach 2: Survey-Based Estimation. Multi-level surveys can estimate complementarity directly:

1. Measure individual life satisfaction (U^0)
2. Measure tribal/community identification (strength of IDN^2)
3. Measure community-level welfare indicators (U^2)
4. Estimate: $U^0 = \alpha + \beta_1 \cdot U^2 + \beta_2 \cdot IDN^2 + \gamma \cdot U^2 \cdot IDN^2 + \epsilon$
5. The interaction term γ estimates cross-level complementarity

Approach 3: Structural Estimation. Given the GTWH model structure, standard structural estimation applies:

$$\mathbf{U}(t+1) = \mathbf{A} \cdot \mathbf{U}(t) + \mathbf{U}(t)^\top \cdot \mathbf{B} \cdot \mathbf{U}(t) + \boldsymbol{\epsilon}(t) \tag{20}$$

With panel data on welfare at multiple levels, the matrices $\mathbf{A}$ and $\mathbf{B}$ (which contain the γ^{cross} terms) can be estimated via:

- Maximum likelihood estimation
- Generalized method of moments

- Bayesian estimation

Approach 4: LLM Monte Carlo. As developed in Appendix N of the main paper, Large Language Model Monte Carlo methods can estimate parameters by simulating agent behavior and calibrating to match observed data patterns.

The Numbers in This Paper. The specific values in Table 2 (e.g., $\gamma_{0,2}^{cross} = +0.40$) are *illustrative estimates* based on:

- Literature on peer effects and community impact
- Plausible magnitudes given documented effect sizes
- Internal consistency checks

They are *not* presented as precise measurements but as reasonable starting points for empirical refinement.

Precedent. Many important economic parameters were used conceptually before precise measurement:

- Elasticity of substitution
- Social discount rate
- Risk aversion coefficient

Measurement difficulty does not invalidate the concept.

Conclusion. Cross-level complementarity is measurable via multiple methods. Current estimates are illustrative, but the research program is empirically tractable.

A.7 Objection A7: "Identity Is Socially Constructed, Not Utility-Maximizing" (Sociologists)

A.7.1 The Objection

Sociologist's Objection

"Your treatment of identity (IDN) as a utility component is reductive. Identity is socially constructed through discourse, power relations, and historical processes—not utility maximization. Reducing identity to economics misses everything important about how identities form and change."

A.7.2 Response

We agree that identity is socially constructed. GTWH does not deny this; it provides an *economic framework* for analyzing constructed identities.

Construction and Utility Are Compatible. Identities are indeed constructed through social processes. But *once constructed*, identities affect behavior in ways that can be modeled economically:

- Identifying as a "miner" affects occupational choices
- Identifying as "French" affects language use, consumption, politics
- Identifying as "vegan" affects food choices

GTWH models these *behavioral effects* of identity, not the construction process itself.

Akerlof and Kranton's Framework. Akerlof and Kranton (2000, 2010) incorporated identity into economics without denying social construction:

$$U_i = U_i(a_i, a_{-i}, I_i) \tag{21}$$

where I_i is identity. They explicitly acknowledge that I_i is shaped by social categories, norms, and history.

GTWH extends this by noting that shared identity ($I_i = I_j$ for $i, j \in$ group) *defines* the group level.

Power and Economics. The objection implies a dichotomy: either power/discourse *or* utility/economics. This is a false dichotomy:

- Power relations shape *which identities are available*
- Economics analyzes *behavior given available identities*
- Both are necessary for complete analysis

GTWH Accommodates Power. The context vector Ψ in GTWH includes institutional and social dimensions:

- Ψ_I: Institutional quality (captures formal power structures)
- Ψ_S: Social trust (captures informal power relations)

These affect which levels are salient and how utilities aggregate.

Empirical Convergence. Sociological and economic approaches increasingly converge:

- DiMaggio (1997): Culture and cognition in economic sociology
- Granovetter (1985): Economic action is embedded in social structures
- Zelizer (2012): Economic sociology incorporates meaning-making

GTWH is compatible with this convergence.

Conclusion. GTWH does not reduce identity to economics. It provides an economic framework for analyzing the *behavioral consequences* of socially constructed identities. The framework is complementary to, not competitive with, sociological analysis of identity construction.

A.8 Objection A8: "The Meta Level Is Utopian Nonsense" (Political Realists)

A.8.1 The Objection

Realist's Objection

"Your 'Meta level' encompassing all sentient beings is utopian philosophy, not serious analysis. Real politics and economics operate at the national level and below. There is no global welfare function, no world government, no mechanism to optimize U^{Meta}. This is irrelevant idealism."

A.8.2 Response

The Meta level serves a specific theoretical function; it is not a policy prescription.

Theoretical Function 1: Upper Bound. The Meta level provides a well-defined *upper bound* for the hierarchy. Without it, the question "Why stop at nations?" has no principled answer.

Theoretical Function 2: Benchmark. U^{Meta} serves as a *normative benchmark*. We can ask:

- How far is current policy from Meta-optimal?
- What is the welfare cost of nationalism (optimizing L6 vs. L8)?
- How much do borders reduce global welfare?

This is analogous to using "perfect competition" as a benchmark even though markets are never perfectly competitive.

Theoretical Function 3: Intergenerational Analysis. Climate change, existential risk, and long-term policy require intergenerational analysis. U^{Meta} provides the framework:

$$U^{Meta} = \int_{t=0}^{\infty} \sum_{i \in S_t} \delta^t \cdot U_t^i \, dt \tag{22}$$

This is not utopian; it is the foundation of Stern (2007), cost-benefit analysis of climate policy, and effective altruism.

Practical Relevance. The Meta level has increasing practical relevance:

- **Climate policy**: Requires global coordination (L8 optimization)
- **Pandemic response**: COVID-19 showed limits of national (L6) optimization
- **AI safety**: Existential risk affects all sentient beings
- **Global health**: GiveWell, Gates Foundation optimize across countries

Not a Policy Prescription. GTWH does not claim we *should* have world government or that national interests don't matter. It provides a framework for analyzing *trade-offs* between levels:

- When should national interests (L6) override global (L8)?
- What is the welfare cost of prioritizing L6?
- Under what conditions is L8 optimization feasible?

These are analytical questions, not ideological commitments.

Conclusion. The Meta level is a theoretical construct with specific analytical functions. It is not utopian idealism but rigorous welfare economics extended to its logical limit—the same extension made by classical utilitarians, climate economists, and effective altruists.

A.9 Objection A9: "Tribal Identity Is Often Harmful" (Psychologists)

A.9.1 The Objection

Psychologist's Objection

"Your theory celebrates tribal identity (L2) as having the strongest cross-level complementarity. But tribal psychology is the source of prejudice, xenophobia, and intergroup conflict. Tajfel's minimal group studies show that *any* group identity triggers discrimination. You're endorsing the dark side of human nature."

A.9.2 Response

This objection confuses description with endorsement. GTWH describes how levels function; it does not claim all level effects are good.

Description vs. Endorsement. The finding that $\gamma_{0,2}^{cross} = +0.40$ is a *descriptive* claim: individual welfare and tribal welfare are strongly complementary. This is consistent with tribal identity having both positive *and* negative consequences:

Positive effects:

- Social support, belonging, meaning
- Cooperation within group
- Mental health benefits of community

Negative effects:

- Prejudice against out-groups
- Intergroup conflict
- Conformity pressure

GTWH accommodates both.

The Negative Effects Appear in the Model. Intergroup conflict appears as *negative spillover* between tribes:

$$\sigma_{ij}^{L2} < 0 \quad \text{(Tribe } i \text{ vs. Tribe } j\text{)} \tag{23}$$

Conflict between in-group and out-group is not ignored—it is modeled as negative horizontal spillover at L2.

The Tribe-Nation Tension. The *negative* cross-level complementarity $\gamma_{2,6}^{cross} = -0.12$ captures exactly the concern: strong tribal identity can *undermine* national cohesion (separatism, sectarianism).

Policy Implication. Recognizing tribal psychology is not endorsing it. The policy question is:

- How to capture the *benefits* of tribal identity (support, meaning)
- While minimizing the *costs* (prejudice, conflict)

GTWH provides the framework for this analysis.

Empirical Balance. Research suggests tribal identity has net positive effects on individual welfare when:

- Out-group hostility is low
- Multiple cross-cutting identities exist
- Superordinate identities (L6, L8) are also active

Roccas and Brewer (2002) document conditions for "social identity complexity" that mitigates negative effects.

Conclusion. GTWH recognizes the power of tribal identity without endorsing its negative manifestations. The negative effects (prejudice, conflict) are explicitly modeled through spillover terms and cross-level tensions. This enables rigorous policy analysis rather than naive celebration.

A.10 Objection A10: "This Is Just Utilitarianism With Extra Steps" (Philosophers)

A.10.1 The Objection

Philosopher's Objection

"Strip away the jargon, and GTWH is just classical utilitarianism: maximize the sum of utilities. Bentham and Mill said this 200 years ago. The 'levels' and 'complementarities' add complexity without changing the fundamental (and problematic) utilitarian framework."

A.10.2 Response

GTWH shares utilitarianism's commitment to welfare as the ultimate criterion, but differs in crucial ways.

Difference 1: Non-Additive Aggregation. Classical utilitarianism:

$$W = \sum_i U_i \quad \text{(additive)} \tag{24}$$

GTWH:

$$W = \sum_i \omega_i U_i + \sum_{i<j} \gamma_{ij} U_i U_j \quad \text{(with interactions)} \tag{25}$$

The interaction term fundamentally changes the analysis:

- Pure utilitarianism is indifferent between $(U_A = 10, U_B = 0)$ and $(U_A = 5, U_B = 5)$
- With $\gamma_{AB} > 0$: $(5, 5)$ produces more total welfare (complementarity bonus)
- This captures the value of *shared* prosperity beyond mere sum

Difference 2: Levels Are Endogenous. Utilitarianism takes individuals as given. GTWH shows that the *relevant units* emerge from identity and collective structure. This answers questions utilitarianism cannot:

- Should we aggregate at family, tribal, or national level?
- How do levels relate to each other?
- Why does community matter beyond individual benefits?

Difference 3: Context-Dependent Salience. Utilitarianism assumes fixed utility functions. GTWH recognizes that *which utilities matter* depends on context:

$$L^*(\Psi) = \arg\max_L [A^L(\Psi)] \tag{26}$$

This explains why the same person acts individualistically in markets but communally in family settings.

Difference 4: Cross-Level Trade-offs. Utilitarianism sums all individual utilities. GTWH reveals that optimizing at one level may *worsen* welfare at another:

$$\gamma_{2,6}^{cross} < 0 \implies \text{Tribe-Nation tension} \tag{27}$$

This is not "utilitarianism with extra steps"; it is a structural enrichment that reveals trade-offs invisible to classical utilitarianism.

Relation to Criticisms of Utilitarianism. Standard objections to utilitarianism are *mitigated* by GTWH:

- **"Ignores distribution"**: Complementarity terms favor shared prosperity over concentrated gains
- **"Ignores relationships"**: Levels explicitly model relational structure
- **"Ignores identity"**: IDN is a core component defining levels
- **"Too demanding"**: Level salience limits relevant scope of optimization

Conclusion. GTWH is a *structural extension* of welfare analysis that addresses key limitations of classical utilitarianism. It is not "utilitarianism with extra steps" but a richer framework that answers questions utilitarianism cannot address.

A.11 Summary: The Defensibility of GTWH

Table 6: Objections and Responses Summary

Objection	**Field**	**Response**
Groups can't have utility	Economics	Functional aggregation, not ontological claim
Levels are arbitrary	Sociology	Levels are discovered via endogeneity principle
Sentience is undefined	Philosophy	Functional criterion; uncertainty manageable
Downward causation is incoherent	Philosophy	Temporal indexing resolves circularity
Self-similarity is ad hoc	Economics	Theoretically grounded, empirically supported
Γ^{cross} is unmeasurable	Empiricism	Multiple measurement approaches exist
Identity is socially constructed	Sociology	Construction and utility are compatible
Meta level is utopian	Political Science	Theoretical benchmark, not policy prescription
Tribal identity is harmful	Psychology	Describes effects; negative effects modeled
Just utilitarianism	Philosophy	Structural enrichment addressing util. limitations

B Comprehensive Glossary of GTWH Concepts

This glossary provides rigorous mathematical definitions and scientific foundations for all key concepts in the General Theory of Welfare Hierarchies. Each entry includes: (1) formal definition, (2) mathematical specification, (3) key literature, and (4) relation to GTWH.

A

Aggregation (Welfare Aggregation). *Definition*: The process by which lower-level utilities combine to form higher-level welfare.

Mathematical Specification:

$$U^L = \underbrace{\sum_{i \in M^L} \omega_i^L \cdot U_i^{L-1}}_{\text{Weighted sum}} + \underbrace{\sum_{i<j} \gamma_{ij}^L \cdot U_i^{L-1} \cdot U_j^{L-1}}_{\text{Complementarity term}} \tag{28}$$

where $\omega_i^L \geq 0$ are aggregation weights satisfying $\sum_i \omega_i^L = 1$, and γ_{ij}^L captures interaction effects between members i and j.

Key Literature: Arrow (1951) showed preference aggregation faces impossibility results; GTWH resolves this by aggregating *welfare* (cardinal) rather than *preferences* (ordinal), and by including complementarity terms.

Relation to GTWH: Aggregation is the upward-directed component of inter-level dynamics (Axiom AL-3).

Altruistic Punishment. *Definition*: Costly punishment of norm violators by individuals who receive no direct material benefit from punishing.

Mathematical Specification:

$$U_i^{punish} = U_i^{base} - c_p \cdot \mathbb{1}_{punish} + \alpha_i \cdot \Delta_{fairness} \tag{29}$$

where $c_p > 0$ is the cost of punishment, $\mathbb{1}_{punish}$ is the punishment indicator, and $\alpha_i \geq 0$ captures individual sensitivity to fairness restoration.

Key Literature: Fehr & Gächter (2002) demonstrate that altruistic punishment sustains cooperation even when costly. Fehr, Fischbacher & Gächter (2002) show this cannot be explained by standard evolutionary theories but is consistent with multilevel selection.

Relation to GTWH: Altruistic punishment is a mechanism sustaining level coherence; it operates at L2 (tribal) level most strongly.

Awareness Function (A^L). *Definition*: The degree to which a given level L is cognitively salient and motivationally active for an individual in context Ψ.

Mathematical Specification:

$$A^L(\Psi) = f\left(IDN_i^L, \Psi_S, \Psi_I, \Psi_E\right) \in [0,1] \tag{30}$$

where IDN_i^L is identity strength at level L, and Ψ_S, Ψ_I, Ψ_E are social, institutional, and environmental context variables.

Key Literature: Akerlof & Kranton (2000) on identity salience; Kahneman (2011) on attention and System 1/System 2 processing.

Relation to GTWH: Determines which level dominates behavior via $L^*(\Psi) = \arg\max_L[A^L(\Psi)]$.

C

Complementarity (Economic Complementarity). *Definition*: A relationship between variables where increasing one raises the marginal value of increasing the other.

Mathematical Specification:

$$\gamma_{ij} = \frac{\partial^2 U}{\partial x_i \partial x_j} > 0 \quad \text{(complements)} \tag{31}$$

$$\gamma_{ij} = \frac{\partial^2 U}{\partial x_i \partial x_j} < 0 \quad \text{(substitutes)} \tag{32}$$

For utility interactions:

$$\gamma_{ij} \cdot U_i \cdot U_j > 0 \implies \text{welfare complementarity} \tag{33}$$

Key Literature: Milgrom & Roberts (1990, 1995) formalized complementarity in organizational economics; Topkis (1998) provided the mathematical foundations via lattice theory.

Relation to GTWH: Complementarity is the mechanism by which group welfare exceeds the sum of individual welfares (the "whole greater than parts" phenomenon).

Conditional Cooperation. *Definition*: A behavioral strategy where an individual cooperates if and only if others cooperate (or are expected to cooperate).

Mathematical Specification:

$$c_i = \begin{cases} 1 & \text{if } \mathbb{E}[\bar{c}_{-i}] \geq \theta_i \\ 0 & \text{otherwise} \end{cases} \tag{34}$$

where $c_i \in \{0,1\}$ is cooperation choice, $\bar{c}_{-i}$ is average cooperation of others, and θ_i is individual threshold.

More generally (continuous version):

$$c_i = \beta_0 + \beta_1 \cdot \mathbb{E}[\bar{c}_{-i}] + \epsilon_i \tag{35}$$

Key Literature: Fischbacher, Gächter & Fehr (2001) document that ∼50% of subjects are conditional cooperators. Fehr & Schurtenberger (2018) show conditional cooperation is the normative foundation of human cooperation.

Relation to GTWH: Conditional cooperation creates positive feedback loops that can sustain high-cooperation equilibria at group levels.

Context Vector (Ψ). *Definition*: A multidimensional vector capturing all environmental, social, and institutional factors that affect behavior and welfare.

Mathematical Specification:

$$\Psi = (\Psi_E, \Psi_S, \Psi_I, \Psi_T, \Psi_C, \Psi_P, \Psi_D, \Psi_N) \in \mathbb{R}^8 \tag{36}$$

where:

- Ψ_E: Economic scarcity/abundance
- Ψ_S: Social trust level
- Ψ_I: Institutional quality
- Ψ_T: Time pressure
- Ψ_C: Cognitive load
- Ψ_P: Physical environment
- Ψ_D: Digital environment
- Ψ_N: Normative clarity

Key Literature: The 8Ψ-Dimensions Framework (BCM 2.0); Ross & Nisbett (1991) on situational determinants of behavior.

Relation to GTWH: Context determines level salience, parameter values (λ, δ), and equilibrium selection.

Cross-Level Complementarity ($\gamma^{cross}_{L,L'}$). *Definition*: The degree to which welfare at level L and welfare at level L' are complements or substitutes in total welfare.

Mathematical Specification:

$$\gamma^{cross}_{L,L'} = \frac{\partial^2 U^{total}}{\partial U^L \partial U^{L'}} \tag{37}$$

Matrix form:

$$\Gamma^{cross} = \begin{pmatrix} 0 & \gamma_{0,1}^{cross} & \gamma_{0,2}^{cross} & \cdots \\ \gamma_{1,0}^{cross} & 0 & \gamma_{1,2}^{cross} & \cdots \\ \gamma_{2,0}^{cross} & \gamma_{2,1}^{cross} & 0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix} \tag{38}$$

Key Literature: Novel to GTWH; extends Milgrom & Roberts (1990) complementarity to cross-level analysis.

Relation to GTWH: Key innovation explaining why some level combinations reinforce each other ($\gamma^{cross} > 0$) while others conflict ($\gamma^{cross} < 0$).

D

Discounting (Temporal Discounting). *Definition*: The reduction in present value of future outcomes.

Mathematical Specification:

Exponential discounting:

$$U_{t=0} = \sum_{t=0}^{\infty} \delta^t \cdot u_t, \quad \delta \in (0,1) \tag{39}$$

Hyperbolic discounting:

$$U_{t=0} = \sum_{t=0}^{\infty} \frac{1}{1 + k \cdot t} \cdot u_t, \quad k > 0 \tag{40}$$

Context-dependent discounting (GTWH):

$$\delta_d^L(\Psi) = \bar{\delta}_d^L \cdot \exp\left(- \sum_k \phi_k \cdot \Psi_k \right) \tag{41}$$

Key Literature: Samuelson (1937) introduced exponential discounting; Laibson (1997) and Frederick, Loewenstein & O'Donoghue (2002) on hyperbolic discounting and present bias.

Relation to GTWH: Discounting operates at all levels; the self-similar structure means δ^L has the same functional form but different parameter values across levels.

Downward Causation. *Definition*: The causal influence of higher-level states on lower-level entities.

Mathematical Specification:

$$U_i^L(t+1) = \bar{U}_i^L + \sum_{L' > L} \beta_{i,L'}^{down} \cdot U^{L'}(t) \cdot \nVdash_{i \in M^{L'}} \tag{42}$$

where $\beta_{i,L'}^{down}$ is the downward causation coefficient from level L' to individual i at level L.

Key Literature: Campbell (1974) introduced the concept in philosophy of biology; Kim (1999) raised the causal exclusion problem; GTWH resolves this via temporal indexing.

Relation to GTWH: Axiom AL-9; explains how macro-level changes (recessions, institutional reforms) affect individual welfare.

E

Endogeneity Principle. *Definition*: The principle that social levels are not exogenously given but emerge from shared identity and collective utility.

Mathematical Specification:

$$\text{Level } L \text{ exists} \iff \exists\, IDN^L \vee \exists\, KNU^L \tag{43}$$

Formally:

$$\mathcal{L} = \{L : \exists\, i, j \in M^L \text{ s.t. } IDN_i^L \cap IDN_j^L \neq \emptyset \vee KNU^L(i,j) > 0\} \tag{44}$$

Key Literature: Novel to GTWH; synthesizes Akerlof & Kranton (2000) on identity with Ostrom (1990) on collective action.

Relation to GTWH: Foundational principle (Axiom AL-2); distinguishes GTWH from approaches that assume levels exogenously.

Equilibrium (Welfare Equilibrium). *Definition*: A state where welfare at all levels is stable under the dynamic system.

Mathematical Specification:

$$\mathbf{U}^* = \mathbf{A} \cdot \mathbf{U}^* + (\mathbf{U}^*)^\top \cdot \mathbf{B} \cdot \mathbf{U}^* \tag{45}$$

Stability condition:

$$\rho(\mathbf{A}) < 1 \tag{46}$$

where $\rho(\cdot)$ denotes the spectral radius.

Key Literature: Standard fixed-point theory; Nash (1950) for game-theoretic equilibrium; Debreu (1959) for general equilibrium.

Relation to GTWH: The complete dynamic system has well-defined equilibria under the stability condition.

G

Gain-Loss Asymmetry (Loss Aversion). *Definition*: The empirical finding that losses loom larger than equivalent gains in subjective valuation.

Mathematical Specification:

$$U = G - \lambda \cdot P, \quad \lambda > 1 \tag{47}$$

where G is gains, P is losses (pains), and λ is the loss aversion coefficient.

Kahneman-Tversky value function:

$$v(x) = \begin{cases} x^{\alpha} & \text{if } x \geq 0 \\ -\lambda \cdot (-x)^{\beta} & \text{if } x < 0 \end{cases} \tag{48}$$

with typical values $\alpha = \beta \approx 0.88$, $\lambda \approx 2.25$.

Key Literature: Kahneman & Tversky (1979) Prospect Theory; Tversky & Kahneman (1991) on loss aversion.

Relation to GTWH: Loss aversion is embedded in the utility function at every level (self-similarity); $\lambda^L(\Psi)$ is context-dependent.

H

Horizontal Spillover. *Definition*: The causal influence of entities at the same level on each other.

Mathematical Specification:

$$U_j^L = \bar{U}_j^L + \sum_{i \neq j} \sigma_{ij}^L \cdot U_i^L \tag{49}$$

where σ_{ij}^L is the spillover coefficient from entity i to entity j at level L.

Matrix form:

$$\Sigma^L = \begin{pmatrix} 0 & \sigma_{12}^L & \cdots \\ \sigma_{21}^L & 0 & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix} \tag{50}$$

Key Literature: Manski (1993) on identification of social effects; Sacerdote (2001) on peer effects; Christakis & Fowler (2007) on network spillovers.

Relation to GTWH: Axiom AL-10; captures how tribes affect neighboring tribes, firms affect competitors, nations affect allies.

I

Identity Utility (IDN). *Definition*: The component of utility derived from self-concept and group membership.

Mathematical Specification:

$$IDN_i = \sum_g w_{ig} \cdot I_{ig} \cdot \left(1 - |a_i - p_g|^2\right) \tag{51}$$

where w_{ig} is weight on group g, I_{ig} is identification strength, a_i is individual's actions, and p_g is group prescriptions.

Key Literature: Akerlof & Kranton (2000, 2010) Economics and Identity; Tajfel & Turner (1979) Social Identity Theory.

Relation to GTWH: IDN defines which levels exist (Endogeneity Principle) and determines level salience.

Inequality Aversion. *Definition*: A preference for equitable outcomes, involving disutility from both advantageous and disadvantageous inequality.

Mathematical Specification (Fehr-Schmidt model):

$$U_i = x_i - \alpha_i \cdot \frac{1}{n-1} \sum_{j \neq i} \max(x_j - x_i, 0) - \beta_i \cdot \frac{1}{n-1} \sum_{j \neq i} \max(x_i - x_j, 0) \tag{52}$$

where $\alpha_i \geq 0$ captures disadvantageous inequality aversion and $\beta_i \in [0, 1)$ captures advantageous inequality aversion, with $\alpha_i \geq \beta_i$.

Key Literature: Fehr & Schmidt (1999) "A Theory of Fairness, Competition, and Cooperation" (QJE); Bolton & Ockenfels (2000) ERC model.

Relation to GTWH: Inequality aversion operates at all levels and creates complementarity in welfare distribution.

K

Collective Utility (KNU). *Definition*: Utility derived from collective outcomes that cannot be reduced to individual utilities alone.

Mathematical Specification:

$$KNU^L = f\left(\sum_i U_i^{L-1}, \sum_{i<j} \gamma_{ij} U_i^{L-1} U_j^{L-1}, X^L\right) \tag{53}$$

where X^L represents collective goods at level L (public goods, shared resources, collective reputation).

Key Literature: Olson (1965) on collective action; Ostrom (1990) on governing the commons; Fehr & Gächter (2000) on cooperation in public goods.

Relation to GTWH: KNU, together with IDN, defines which levels exist (Endogeneity Principle).

L

Level (Social Level). *Definition*: A stratum of social organization defined by shared identity or collective utility.

Mathematical Specification:

$$L \in \{0, 1, 2, 3, ..., \infty\} \tag{54}$$

with ordering:

$$L_1 < L_2 \iff M^{L_1} \subseteq M^{L_2} \tag{55}$$

where M^L is the membership set of level L.

Standard levels in GTWH:

$L = 0$	Individual
$L = 1$	Primary group (family)
$L = 2$	Tribe/Community
$L = 3$	Organization
$L = 4$	Network
$L = 5$	Region
$L = 6$	Nation
$L = 7$	Supranational
$L = 8$	Species
$L = \infty$	Meta (all sentient, all time)

Key Literature: Levels are endogenously derived in GTWH; cf. Simon (1962) on hierarchy; Dunbar (1992) on cognitive constraints on group size.

Relation to GTWH: Levels are the fundamental units of analysis; the Endogeneity Principle determines which levels exist in any given context.

Level Salience (L^*). *Definition*: The level that is most cognitively and motivationally active in a given context.

Mathematical Specification:

$$L^*(\Psi) = \arg\max_L \left[A^L(\Psi)\right] \tag{56}$$

With ties broken by:

$$L^*(\Psi) = \arg\max_L \left[A^L(\Psi) + \epsilon \cdot L\right] \tag{57}$$

where $\epsilon \to 0^+$ favors lower (more proximate) levels.

Key Literature: Construal Level Theory (Trope & Liberman, 2010); Identity salience (Stryker, 1980).

Relation to GTWH: Determines which level's utility function governs behavior in a given context.

M

Master Equation (GTWH Dynamic). *Definition*: The complete dynamic equation governing welfare evolution across all levels.

Mathematical Specification:

$$\boxed{U_i^L(t+1) = \underbrace{\sum_j \omega_{ij}^L U_j^{L-1}(t)}_{\text{Upward}} + \underbrace{\sum_{L'>L} \beta_{i,L'}^{down} U^{L'}(t)}_{\text{Downward}} + \underbrace{\sum_{k\neq i} \sigma_{ik}^L U_k^L(t)}_{\text{Horizontal}} + \underbrace{\sum_{L'\neq L} \gamma_{L,L'}^{cross} U^{L'}(t)}_{\text{Cross-Level}}} \tag{58}$$

Matrix form:

$$\mathbf{U}(t+1) = \mathbf{A} \cdot \mathbf{U}(t) + \mathbf{U}(t)^\top \cdot \mathbf{B} \cdot \mathbf{U}(t) \tag{59}$$

Key Literature: Novel to GTWH; synthesizes aggregation theory, downward causation, spillover effects, and complementarity.

Relation to GTWH: The complete specification of inter-level dynamics.

Meta Level ($L = \infty$). *Definition*: The ultimate scope of welfare analysis encompassing all sentient beings across all time.

Mathematical Specification:

$$U^{Meta} = \int_{t=0}^{\infty} \sum_{i \in S_t} \delta^t \cdot U_t^i \, dt \tag{60}$$

where S_t is the set of all sentient beings at time t.

With uncertainty about sentience:

$$U^{Meta} = \int_{t=0}^{\infty} \sum_i P(\text{sentient}_i) \cdot \delta^t \cdot U_t^i \, dt \tag{61}$$

Key Literature: Parfit (1984) on personal identity and future generations; Chalmers (1996) on consciousness; Singer (1981) on the expanding circle.

Relation to GTWH: Provides the well-defined upper bound for the level hierarchy (Axiom AL-1).

Multilevel Selection. *Definition*: The evolutionary process where selection operates at multiple levels (genes, individuals, groups) simultaneously.

Mathematical Specification (Price equation):

$$\Delta\bar{z} = \underbrace{\mathrm{Cov}(w_i, z_i)}_{\text{Individual selection}} + \underbrace{\mathbb{E}[w_i \Delta z_i]}_{\text{Transmission bias}} \tag{62}$$

Extended to group selection:

$$\Delta\bar{z} = \underbrace{\mathrm{Cov}(W_g, \bar{z}_g)}_{\text{Between-group}} + \underbrace{\mathbb{E}[\mathrm{Cov}_g(w_i, z_i)]}_{\text{Within-group}} \tag{63}$$

Key Literature: Price (1970, 1972); Sober & Wilson (1998) "Unto Others"; Fehr, Fischbacher & Gächter (2002) show strong reciprocity is consistent with multilevel selection.

Relation to GTWH: Provides the evolutionary foundation for why welfare hierarchies exist and persist.

R

Recursive Structure (Self-Similarity). *Definition*: The property that the same mathematical structure applies at every level of the hierarchy.

Mathematical Specification:

$$U^L_{d,t}(\Psi) = \left(\delta^L_d(\Psi)\right)^t \cdot \left[G^L_{d,t} - \lambda^L_d(\Psi) \cdot P^L_{d,t}\right] \tag{64}$$

This formula holds for $L \in \{0, 1, 2, ..., \infty\}$ with level-specific parameters.

Key Literature: Self-similarity is common in complex systems (Mandelbrot, 1982); GTWH applies it to social organization. Schelling (1978) shows how micro and macro share structural properties.

Relation to GTWH: Axiom AL-3; enables tractable analysis across all levels with consistent framework.

S

Sentience. *Definition*: The capacity for subjective experience; the property of having "something it is like" to be that entity.

Mathematical Specification:

Functional criterion:

$$\text{Sentient}_i \iff \exists \text{ behavioral markers } B_i \tag{65}$$

Probabilistic treatment:

$$P(\text{sentient}_i) \in [0, 1] \tag{66}$$

Key Literature: Nagel (1974) "What is it like to be a bat?"; Chalmers (1996) on the "hard problem" of consciousness; Singer (1975) on sentience as the criterion for moral consideration.

Relation to GTWH: Sentience defines the base case ($L = 0$) and the ultimate scope (Meta level) of welfare analysis.

Social Preferences. *Definition*: Preferences that include concern for others' outcomes, fairness, and reciprocity beyond pure self-interest.

Mathematical Specification:

$$U_i = u_i(x_i) + \sum_j \alpha_{ij} \cdot u_j(x_j) + \beta_i \cdot F(\mathbf{x}) \tag{67}$$

where α_{ij} captures other-regarding preferences and $F(\mathbf{x})$ captures fairness/inequality concerns.

Key Literature: Fehr & Schmidt (1999) inequality aversion; Charness & Rabin (2002) on social welfare preferences; Fehr & Fischbacher (2002) on why social preferences matter.

Relation to GTWH: Social preferences create the complementarity terms that make group welfare exceed summed individual welfare.

Strong Reciprocity. *Definition*: A behavioral propensity to cooperate with cooperators and punish defectors, even at personal cost and without expectation of future benefit.

Mathematical Specification:

$$a_i = \begin{cases} \text{cooperate} & \text{if partner cooperated} \\ \text{punish at cost } c & \text{if partner defected} \end{cases} \tag{68}$$

Evolutionary condition:

$$\text{Strong reciprocity evolves} \iff \text{Multilevel selection operates} \tag{69}$$

Key Literature: Fehr, Fischbacher & Gächter (2002) coined the term and showed it cannot be explained by kin selection, reciprocal altruism, indirect reciprocity, or costly signaling—but is consistent with multilevel selection and cultural evolution.

Relation to GTWH: Strong reciprocity is the behavioral mechanism that sustains cooperation at all levels; it explains why levels are stable.

Super-Additive Cooperation. *Definition*: The phenomenon where combining repeated interactions with intergroup competition produces cooperation greater than either mechanism alone.

Mathematical Specification:

$$C_{combined} > C_{repeated} + C_{competition} \tag{70}$$

The "super-additive" effect:

$$\Delta C = C_{combined} - (C_{repeated} + C_{competition}) > 0 \tag{71}$$

Key Literature: Efferson, Bernhard, Fischbacher & Fehr (2024) in *Nature* show that neither repeated interactions nor intergroup competition alone reliably sustains cooperation, but their combination does.

Relation to GTWH: Provides direct empirical support for cross-level complementarity ($\gamma^{cross} > 0$); multiple levels must interact.

U

Upward Aggregation. *Definition*: The process by which lower-level utilities combine to form higher-level welfare (see Aggregation).

Mathematical Specification:

$$U^L = \sum_{i \in M^L} \omega_i^L \cdot U_i^{L-1} + \sum_{i<j} \gamma_{ij}^L \cdot U_i^{L-1} \cdot U_j^{L-1} \tag{72}$$

Relation to GTWH: The upward-directed component of inter-level dynamics.

Utility Function (Level-Specific). *Definition*: The mathematical representation of welfare at a given level.

Mathematical Specification:

$$U_{d,t}^L(\Psi) = \left(\delta_d^L(\Psi)\right)^t \cdot \left[G_{d,t}^L - \lambda_d^L(\Psi) \cdot P_{d,t}^L\right] \tag{73}$$

Components:

- $\delta_d^L(\Psi)$: Level- and context-specific discount factor
- $G_{d,t}^L$: Gains in domain d at time t
- $\lambda_d^L(\Psi)$: Loss aversion coefficient

- $P^L_{d,t}$: Pains/losses in domain d at time t

Key Literature: Integrates prospect theory (Kahneman & Tversky, 1979), discounted utility (Samuelson, 1937), and domain-specificity.

Relation to GTWH: The fundamental building block; self-similar across all levels.

W

Welfare Hierarchy. *Definition*: The complete structure of nested social levels with their associated welfare functions and inter-level dynamics.

Mathematical Specification:

$$\mathcal{H} = \left\{(L, U^L, M^L, \omega^L, \gamma^L, \beta^{down}_L, \sigma^L, \gamma^{cross}_L) : L \in \mathcal{L}\right\} \tag{74}$$

where $\mathcal{L}$ is the set of levels defined by the Endogeneity Principle.

Key Literature: Novel to GTWH; synthesizes multi-level analysis (Coleman, 1990), identity economics (Akerlof & Kranton, 2000), and complementarity theory (Milgrom & Roberts, 1990).

Relation to GTWH: The central object of analysis in the General Theory of Welfare Hierarchies.

Summary of Key Equations

Table 7: GTWH Key Equations Summary

Concept	**Equation**
Utility Function	$U^L_{d,t}(\Psi) = \delta^t \cdot [G - \lambda \cdot P]$
Aggregation	$U^L = \sum_i \omega_i U_i^{L-1} + \sum_{i<j} \gamma_{ij} U_i^{L-1} U_j^{L-1}$
Downward Causation	$U^L_i = \bar{U} + \sum_{L'>L} \beta^{down}_{L'} U^{L'}$
Horizontal Spillover	$U^L_j = \bar{U} + \sum_{i \neq j} \sigma_{ij} U^L_i$
Cross-Level Complementarity	$\gamma^{cross}_{L,L'} = \partial^2 U^{total} / \partial U^L \partial U^{L'}$
Level Salience	$L^*(\Psi) = \arg\max_L [A^L(\Psi)]$
Endogeneity Principle	Level L exists $\iff \exists IDN^L \vee \exists KNU^L$
Meta Level	$U^{Meta} = \int_0^\infty \sum_{i \in S_t} \delta^t U^i_t \, dt$
Master Equation	$U^L(t+1) =$ Upward + Downward + Horizontal + Cross-Level